

Imines and Enamines

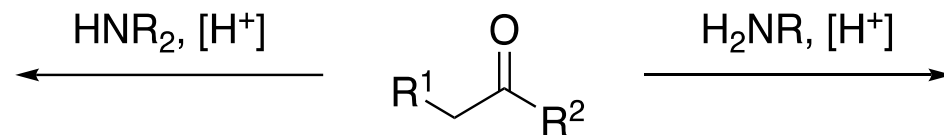
Organic Chemistry II

CHEM 2312

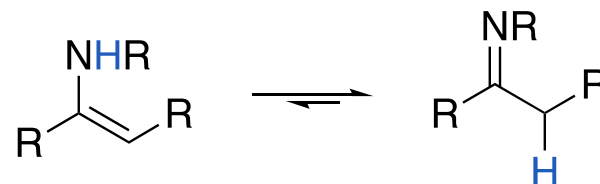
Dr. Michael Evans

1. Recall methods for the preparation of imines, enamines, and iminium ions from ketohydes.
2. Apply imine-enamine tautomerism; draw the enamine tautomer of a given imine and *vice versa*.
3. Predict the products of Mannich reactions of amines, non-enolizable ketohydes, and nucleophilic ketohydes.
4. Apply the Mannich reaction in synthesis.
5. Predict the products of Stork enamine reactions, including conjugate additions to unsaturated carbonyl compounds, acylation, and alkylation of metallocenamines.
6. Apply Stork enamine reactions in synthesis.
7. Recognize and apply enamine- and imine-like structures in heteroarenes and substituted benzenes.
8. Describe biochemical mechanisms involving pyridoxal phosphate, an imine-forming cofactor.

- Recall that treating a ketone or aldehyde with an amine leads either to an **imine** (1° amine) or an **enamine** (2° amine)
- Imine and enamine groups also show up within larger structures such as heteroaromatics



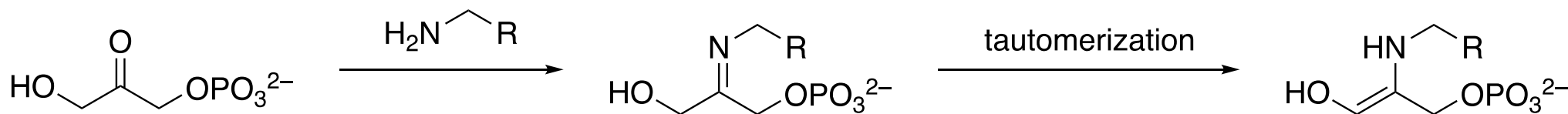
- Imines and enamines (containing at least one N-H) are **tautomers**
- The imine tautomer is typically heavily favored, although enamines can be important in biochemical reactions



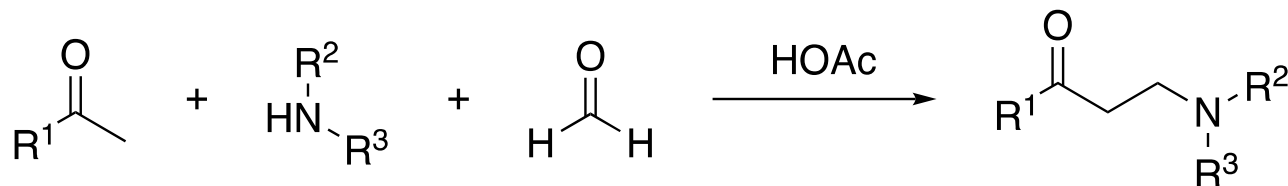
- Protonation of an imine or enamine yields an iminium ion intermediate



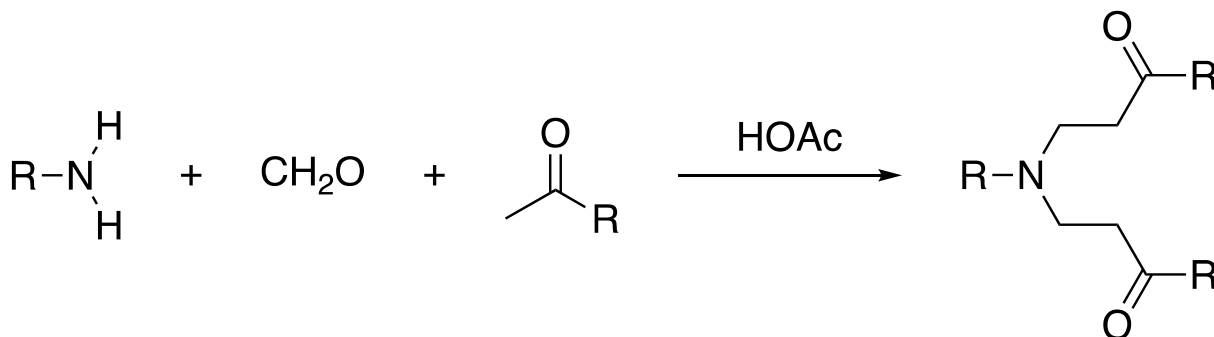
- Class I aldolase enzymes use a lysine-linked enamine as the nucleophilic partner in the aldol reaction



- The **Mannich reaction** combines iminium formation with aldol-type reactivity under neutral or acidic conditions to produce β -amino carbonyl products
- Formaldehyde, benzophenone, and other non-enolizable electrophiles are used

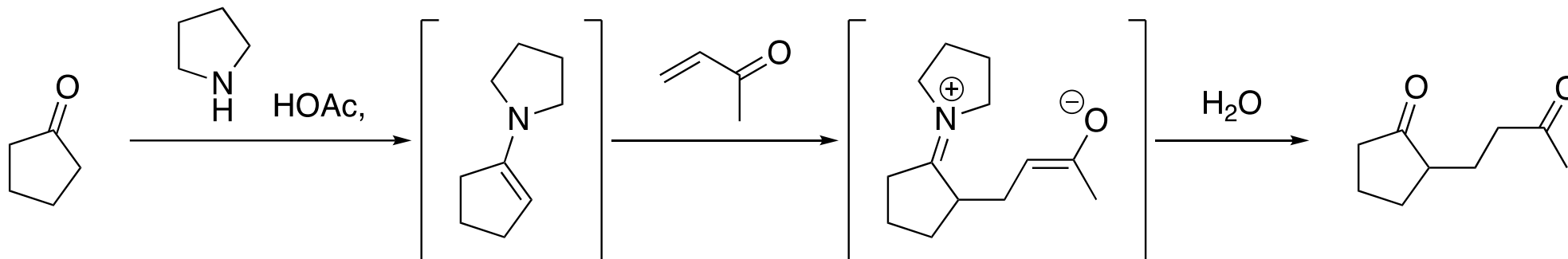


- Primary amines (H₂NR) can undergo two condensations to yield tertiary amines

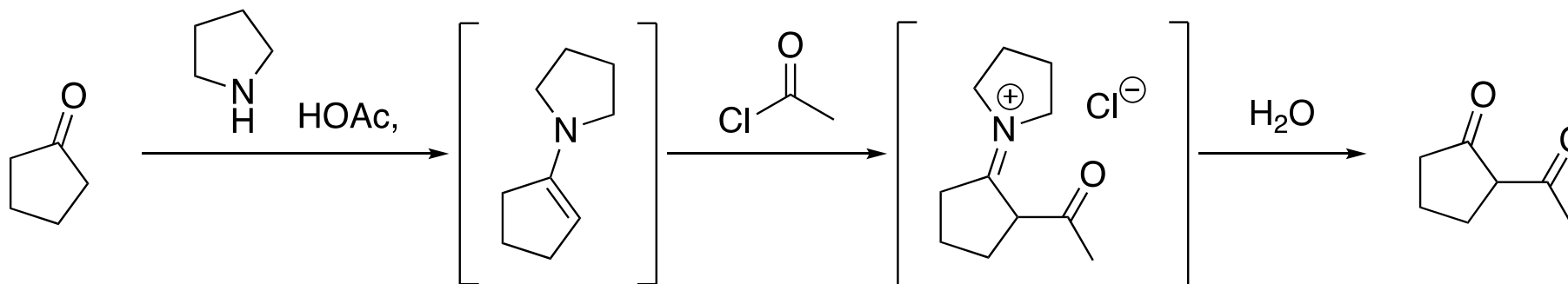


21.6 Stork Enamine Reactions | 1 of 2

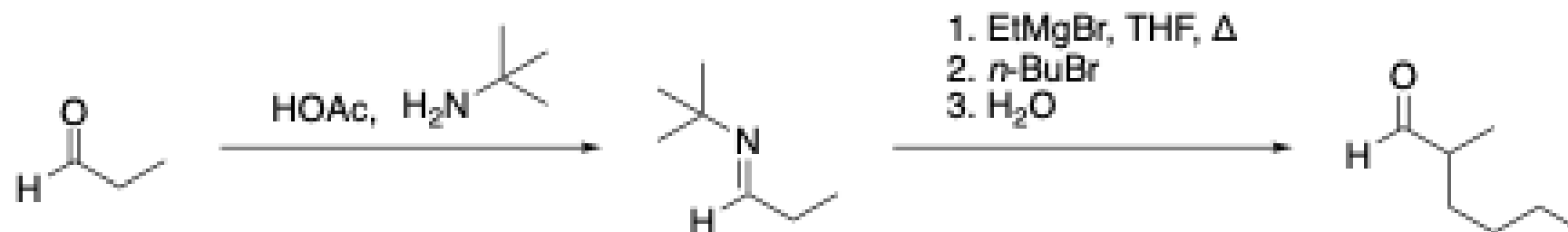
- Recall that simple enolates give mixtures of 1,2- and 1,4-addition
- Enamines engage in selective 1,4-addition with unsaturated carbonyls, a reaction called **Stork alkylation**



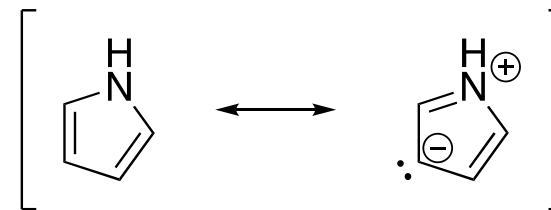
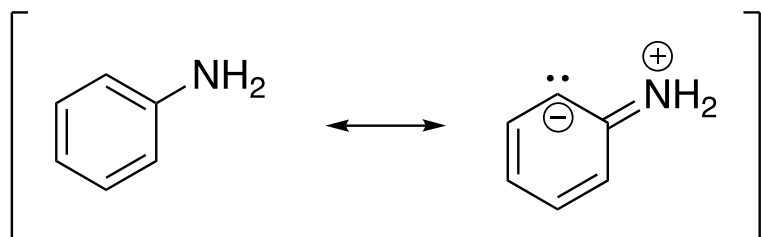
- Enamines can also be acylated at carbon using acyl chlorides (**Stork acylation**)



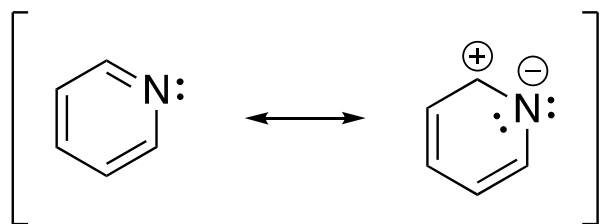
- Enamine anions (**metalloenamines**), generated via deprotonation of imines with ethyl Grignard, can be alkylated using alkyl halides



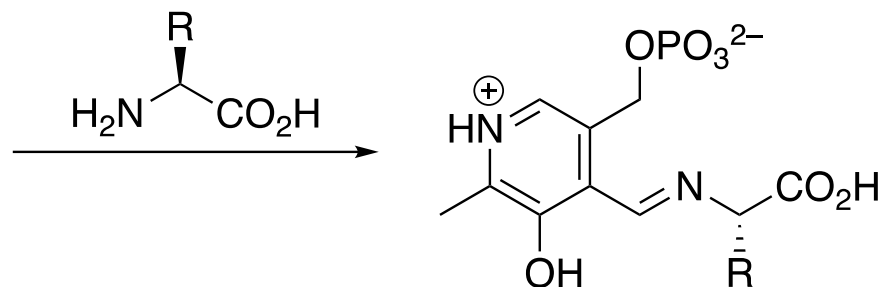
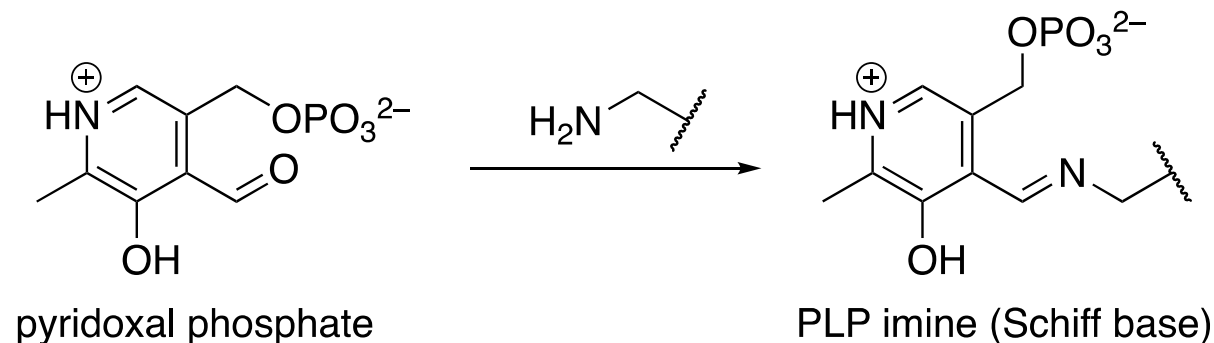
- Anilines contain an enamine-like structure, which explains their extreme reactivity in electrophilic substitutions
- Pyrrole also contains built-in enamine groups



- Pyridines and pyridiniums contain built-in imine and iminium groups
- Pyridines react sluggishly in EAS and can undergo nucleophilic aromatic substitutions

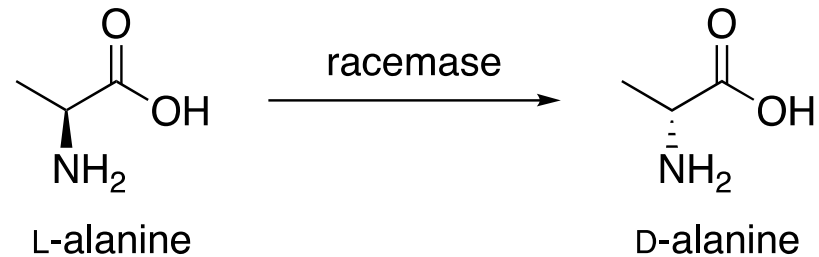


- The cofactor **pyridoxal phosphate (PLP)**, an oxidized form of vitamin B6, forms imines with lysine side chains in enzymatic active sites and with the amino groups in amino acids

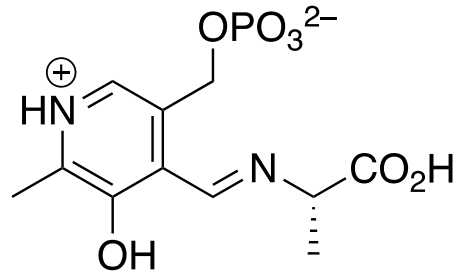


- How is the acidity of the α carbon changed by condensation of the amino acid with PLP?

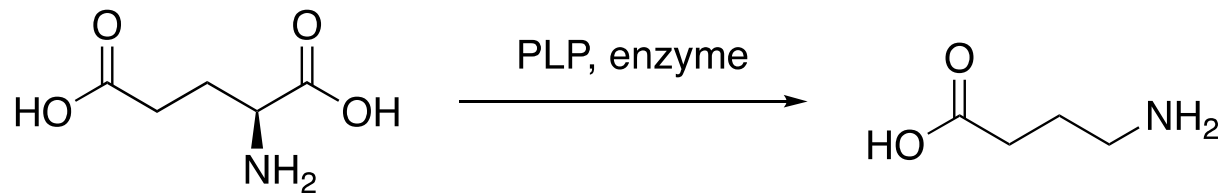
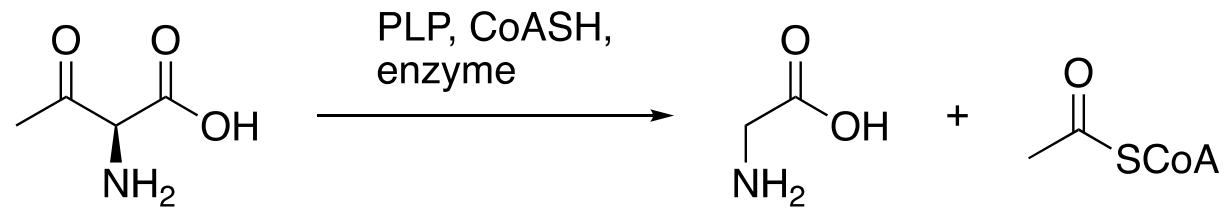
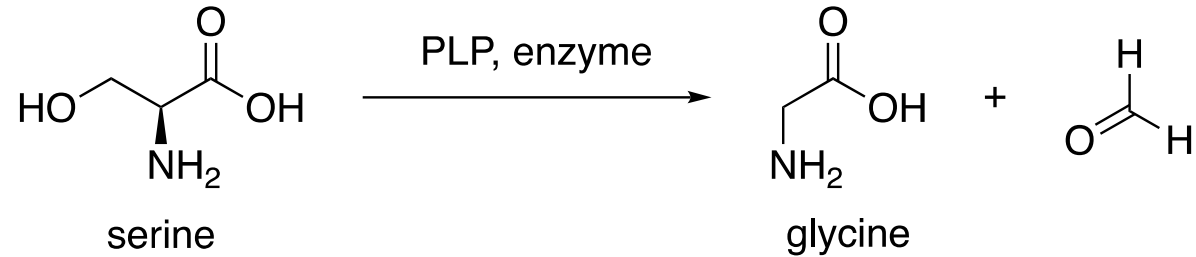
- PLP-dependent racemase enzymes catalyze a change in configuration at the α carbons of amino acids
- The effectiveness of catalysis is a direct result of acidification of the α carbon



- After imine formation, the change in configuration occurs via two proton transfers



- PLP also assists in retro-aldol, retro-Claisen, and decarboxylation reactions of α -amino acids or esters
- All of these involve the formation of an imine in the mechanism



- PLP assists in the biosynthesis of cysteine from serine and acetyl coenzyme A
- Acylation of the serine hydroxyl group is followed by imine formation, elimination, and nucleophilic addition of HS⁻

